

PUPIL WORKSHEET

# Run and judge a fair test

Autumn 1 · Week 5 · Lesson B

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Carry out or direct a controlled two-surface comparison, record three trials where practicable and judge whether evidence answers the question.

Use Monday's plan. Compare two surfaces with three releases each if practicable. Choose an actual practical/directing route or the clearly labelled model-data alternative. Keep every original result and label retries separately.

## 1. Confirm the plan

Question: Does the run-out surface affect stopping distance? Name two available surfaces and check your CHANGE/KEEP/MEASURE plan with the adult before touching the kit.

| Plan item               | Our agreed choice |
|-------------------------|-------------------|
| Surface A and surface B |                   |
| CHANGE                  |                   |
| KEEP THE SAME           |                   |
| MEASURE, with unit      |                   |
| My role                 |                   |

## 2. Record all three trials

*Write, draw, point or dictate as agreed with your teacher.*

Release without a push. Measure from the same zero to the same trolley edge after stopping. Repeat the procedure up to three times per surface. If a run was not performed, say so; never enter a made-up zero.

- Original result kept for checking: \_\_\_\_.
- Method issue or uncertainty: \_\_\_\_.
- Separate retry, if performed: \_\_\_\_; reason: \_\_\_\_.

| Named surface | Trial 1 / cm | Trial 2 / cm | Trial 3 / cm |
|---------------|--------------|--------------|--------------|
| A:            |              |              |              |
| B:            |              |              |              |

### 3. Alternative: model-data analysis

*Write, draw, point or dictate as agreed with your teacher.*

Use only as an alternative to the practical. The source provides this invented teaching set. Explain what the data show and which controls must be true before trusting the comparison.

- Compare A and B using values: \_\_\_\_.
- Two necessary controls: \_\_\_\_ and \_\_\_\_.
- This is data interpretation rather than a practical I performed because \_\_\_\_.

| Model surface | Trial 1 / cm | Trial 2 / cm | Trial 3 / cm |
|---------------|--------------|--------------|--------------|
| A             | 48           | 50           | 49           |
| B             | 27           | 29           | 28           |

## 4. Judge and explain the evidence

*Write, draw, point or dictate as agreed with your teacher.*

Use actual or clearly labelled model results. Inspect an unusual run; note any specific method issue. Write a conclusion with numbers and one limitation. Optional stretch: calculate a mean only for three comparable numeric values.

- My evidence supports \_\_\_\_ because \_\_\_\_.
  - One uncertainty or control to improve: \_\_\_\_.
  - Optional mean A:  $(\text{ } + \text{ } + \text{ }) \div 3 = \text{ } \text{ cm.}$
  - Optional mean B:  $(\text{ } + \text{ } + \text{ }) \div 3 = \text{ } \text{ cm.}$
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## 5. Feedback and exit

*Write, draw, point or dictate as agreed with your teacher.*

Offer the adult one suggestion about testing, waiting or recording, or pass privately. Choose one exit question. Keep your plan with results and conclusion, then return equipment safely.

- What helped, or what could change?
  - Supported: how many things did we deliberately change?
  - Standard: name one control from today's test.
  - Stretch: why are repeats more useful than one result?
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CHOOSE A RESPONSE ROUTE

# Ways to show your understanding

*Choose the route agreed with your teacher. You do not need to complete every route.*

## Supported

Choose CHANGE/KEEP/MEASURE labels with the adult. Direct three releases on one surface and compare an adult-recorded second surface. Dictate values and one conclusion.

## Standard

Plan controls, record up to three trials for each surface in cm and use repeated evidence to answer the question.

## Stretch

Explain an unusual result or limitation. Calculate means where appropriate, keep raw data and separate retries, and explain why a mean cannot repair an unfair method.

Keep the plan, all original data, any separate retry and conclusion together. Next current Week 6 explores Earth and the planets.